Write a short report (max 2 pages) on your results and the challenges you faced in this pipeline. Include figures of your segmentation outcome and the resulting FEM model. As usual, be critical of your results. As for the challenges, how can these be overcome for your project?

- After painting each view, there was troubleshooting to get the 3d image to open. Unable to reproduce procedure as I wasn't able to track what made the 3d render show up initially.
- When resuming slicer3d program after closing, I was unsure if I was saving my work and when trying to reopen the saved .nrrd file, it wasn't where I was hoping to be left off making me unable to provide screenshots of my model.
- Another problem I had was cleaning up the model. I was able to apply to apply the median filter and joint smoothing but using the paint tool was too intensive for my computer to compute and apply. (Time to clean up. These steps may take time depending on your computer. First use the smoothing tool to apply a median filter and joint smoothing (set kernel size 0.17). Next we want to close some holes using the Closing option (set Kernel size to 8.00 mm). Finally, you can use the paint tool again (but now select Edit in 3D views) to close up any remaining gaps you see.)
- I redid the tutorial because my first paint job for selecting the bones produced an unrecogonizable 3d model that was unusable. It was probably user error of the mask selecting more than just the bones so when painting for the individual bones, it selected more than necessary but upon second try, it worked as anticipated.

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BME 60D
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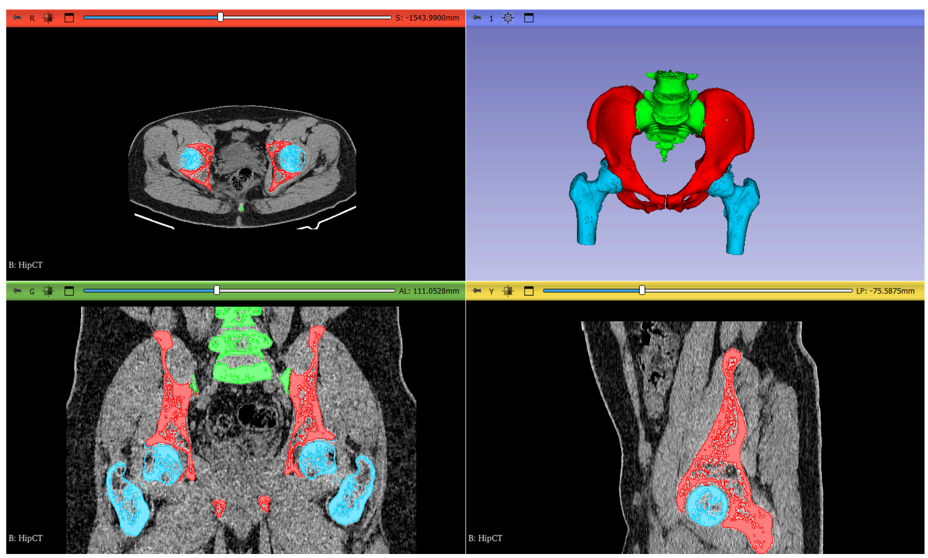
Tutorial 4 - Real Geometries Report

This project was my first hands-on attempt at constructing a realistic anatomical model from medical imaging data. My experience highlighted both the potential of the 3D Slicer-to-Ansys pipeline and the challenges that come with working with real CT datasets and computational tools.

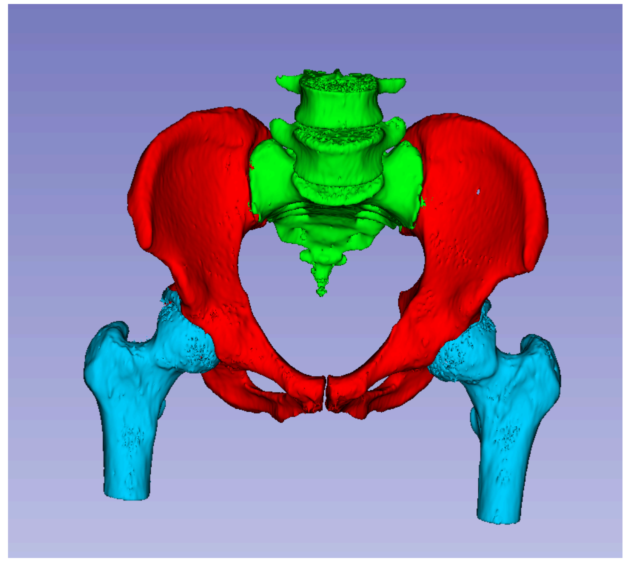
The first step involved loading the CT scan into 3D Slicer and applying the "CT-Abdomen" preset to improve bone contrast. Segment creation began smoothly—I added four segments (mask, femur, sacrum, and hip) and used the Threshold tool to isolate the bone regions. Painting the bones across the red, yellow, and green slice views worked well in theory, but practically it required precision and patience. My first attempt led to a distorted, unrecognizable 3D model. Upon reflection, I realized this was likely due to my initial threshold including non-bone structures, which caused the painted outlines to capture unintended regions.



Redoing the process more carefully a second time, especially reselecting a tighter bone mask before painting, resulted in a much more accurate 3D output. The "Grow from Seeds" function worked well the second time, correctly propagating the painted segments throughout the scan slices.

Once the initial painting and segmentation were complete, I attempted to view the 3D rendering of the model. This step proved inconsistent. I initially got the 3D view to work, but after restarting Slicer, I couldn't reliably reproduce the steps that enabled it. This made troubleshooting difficult, especially without clear documentation or feedback from the software about what was going wrong.

Saving and reopening the project was another issue. After closing Slicer and reopening the .nrrd file, I didn't return to the state I expected. It wasn't clear whether I had saved the right data or how to reload the segmented model. I eventually found my file>recent>C:/Users/brian/Documents/2025-04-28-Scene.mrml.



The most frustrating challenge came during model cleanup. While applying the smoothing filters worked, trying to use the Paint tool in the 3D view for final edits caused my system to freeze. The operations were too computationally demanding for my computer. Despite this, I was still able to use earlier cleanup tools like smoothing and closing to produce a reasonably clean model.

This project gave me insight into the level of attention and computing power required when working with real medical imaging data. Small mistakes in thresholding early on can snowball into unusable models. Likewise, understanding how segmentation tools interact (e.g., mask vs. individual segments) is essential to success.

The 3D Slicer interface, while powerful, has a steep learning curve. Clearer file-saving practices and better visualization controls would greatly enhance usability for beginners. On a technical level, I learned that software like this can quickly outpace system resources, especially during 3D editing or smoothing.

Redoing the tutorial helped me see where I went wrong the first time, and it showed me the importance of iteration and troubleshooting in bioengineering workflows. Even though I didn't end up with perfect results, I gained practical skills in segmentation, CT data handling, and workflow navigation that will be valuable for future modeling projects.